SESSION 1

OUR IDEA=Trustlens-An AI-powered physical-world cybersecurity platform that verifies digital interfaces before user interaction.

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1. Comparing Machine Learning and Deep Learning Models for QR Code Phishing Detection

- **Authors:** Yahya M. Tashtoush, Mahar Khashim, and Nasser H. Alsaedi
- **Publication year:** 2025
- **Publication venue:** *2025 International Conference on Cybersecurity and AI-Based Systems (Cyber-AI)*, pages 291–297.
- **Relevance to TrustLens:** This paper compares traditional machine learning approaches that analyse decoded URLs with deep learning approaches that analyse QR code images. It is relevant to TrustLens because it helps us understand how AI techniques can identify potentially malicious QR codes.

2. TrustQR: A New Technique for the Detection of Phishing Attacks on QR Code

- **Authors:** Ahmad Y. Alnajjar, Selvakumar Manickam, Mohammed Anbar, Samer Al-Saleem, and one additional co-author.
- **Publication year:** 2016
- **Publication venue:** *Advanced Science Letters*, Volume 22, Issue 10, pages 2905–2909.
- **Relevance to TrustLens:** This paper introduces a technique that uses QR-code characteristics and URL features to detect phishing attacks. It provides a foundation for our project's QR-code verification and malicious-link detection components.

3.phishing Detection in Advanced QR Code Attacks: Challenges and AI-Driven Solutions (2025)
Authors: Aayush Trivedi, Krishnappa Jangal, and Rashi Gupta.
Venue: *International Journal for Research in Applied Science and Engineering Technology (IJRASET)*, Volume 13, Issue 1, pages 479–482.
This paper discusses the challenges of QR-code phishing and explores AI-driven detection approaches. It helps us understand the limitations of existing security methods and potential ways to improve detection.

4.**Detection of Phishing Link and QR Code of UPI Transaction using Machine Learning**

- **Authors:** Gangisetty Raj Charan and K. Deepa Thilak.

- **Publication year:** 2023.
- **Publication venue:** *2023 3rd International Conference on Innovative Mechanisms for Industry Applications (ICIMIA)*, IEEE conference proceedings, pages 658–663.
- **Relevance to TrustLens:** This paper focuses on detecting phishing links and QR codes associated with UPI transactions using machine learning. It is relevant to TrustLens because our platform aims to identify potential threats in QR-based interactions, particularly payment-related use cases.

5. **QR Code Phishing Detection: A Multi-Modal Ensemble Learning Framework Combining URL and Image Analysis**

- **Authors:** Nadeline Aprillini, Jessica Seren Sutrisno, Jocelyn Marcella, and Franz Adeta Junior.
- **Publication year:** 2025.
- **Publication venue:** *2025 5th International Conference of Science and Information Technology in Smart Administration (ICSINTESA)*, pages 387–391.
- **Relevance to TrustLens:** This paper combines URL analysis and QR-image analysis using an ensemble learning approach. It is relevant because TrustLens also aims to examine digital destination information and physical-interface evidence rather than relying on only one security signal.

6. **A systematic literature review on deep learning approaches for QR code-based phishing (quishing) detection: emerging attack vectors, multimodal feature fusion and open research challenges**

- **Author:** Kritika
- **Publication year:** 2026
- **Publication venue:** *Journal of Cyber Security Technology*, Volume 10, Issue 1.
- **DOI:** 10.1080/23742917.2026.2696261
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Relevance to TrustLens: This review examines deep learning techniques for detecting QR-code phishing, including image analysis, URL classification, and multimodal approaches. It also discusses limitations such as dataset generalisation, adversarial attacks, and explainability. These findings can help us identify limitations in existing QR security methods and guide TrustLens's design.

7. **QRphish: An Automated QR Code Phishing Detection Approach**

- **Authors:** A. S. Mashaleh, Hind Mousa Al-Hamadeen, and Amjad Gawanmeh
- **Publication year:** 2026
- **Publication venue:** *2026 2nd International Conference on Computational Intelligence Approaches and Applications (ICCIAA)*, IEEE.
- **DOI:** 10.1109/ICCIAA68481.2026.11543700

Relevance to TrustLens: This paper presents an automated approach to detecting phishing attacks involving QR codes. It is relevant to our QR verification component and can help us compare existing automated detection techniques with our proposed combination of URL checks and registered-interface verification.

8. **Quantum Machine Learning in Cybersecurity: A Hybrid Convolutional Neural Network with Skip Connections for QR Code Phishing Detection**

- **Authors: Could not be verified.**
- **Publication year: Could not be verified.**
- **Publication venue: Could not be verified.**

I could not locate a reliable publication record matching this exact title. I found related research on quantum machine learning for phishing detection, but that does not establish that this specific paper exists.

Relevance to TrustLens: If this paper is genuine, it may be useful for understanding experimental quantum machine learning approaches to QR-code phishing detection. However, verify the exact title and bibliographic information on Google Scholar, IEEE Xplore, ACM Digital Library, or the publisher's website before including it as a confirmed publication.